

Khalid Zobaid Adnan

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RESEARCH SUMMARY

Ph.D. candidate in mechanical engineering working on computational interfacial thermal transport. I use density-functional theory, machine-learning interatomic potentials, and nonequilibrium molecular dynamics on multi-GPU HPC systems to predict thermal boundary conductance in wide- and ultrawide-bandgap semiconductor heterostructures. Five peer-reviewed papers (four first-author) in *Physical Review B*, *Physical Review Applied*, *Applied Physics Letters*, and the *International Journal of Heat and Mass Transfer*. Central result: thermal boundary conductance is not a property of a single interface, but depends on the nearby environment.

EDUCATION

Ph.D., Mechanical Engineering Jan 2023 – present

University of Utah, Salt Lake City, UT

Proposed thesis: interfacial thermal transport and the interplay between successive interfaces in solid–solid heterostructures. Advisor: Tianli Feng.

Expected graduation: summer 2027.

MSc, Mechanical Engineering Jan 2023 – Dec 2025

University of Utah, Salt Lake City, UT

GPA 4.0/4.0. Summa Cum Laude (top 1% among the graduating class of a five-year period).

BSc, Mechanical Engineering 2016 – 2021

Bangladesh University of Engineering and Technology (BUET), Dhaka

CGPA 3.53/4.00.

RESEARCH EXPERIENCE

Graduate Research Assistant Spring 2023 – present

Feng Research Group, University of Utah

- Predict interfacial thermal transport across Si/Ge, metal (Al, Zr, Mo, Au)/diamond, Si/diamond, GaN/AlN, GaN/AlGaIn, and GaN/diamond interfaces relevant to thermal management in power and RF electronics.
- Develop machine-learning interatomic potentials (DeePMD, MTP) from ab initio molecular dynamics, enabling nonequilibrium molecular dynamics in cells far beyond the reach of direct DFT.
- Run large-scale DFT and molecular-dynamics campaigns on multi-GPU HPC clusters under SLURM, with DFT systems of 300–500 atoms.
- Implemented and quantified the quantum correction to the classical diffuse mismatch model, placing molecular dynamics, analytical models, and experiment on a common footing.
- Built a finite-element silicon-on-diamond device model translating interfacial conductance into hot-spot temperature rise and device reliability.
- Collaborations: U.S. Army DEVCOM Army Research Laboratory (metal/diamond); Sandia National Laboratories (GaN/diamond, in progress).
- Supported by NSF CAREER award CBET-2337749; computing through CHPC (University of Utah) and Bridges-2 (Pittsburgh Supercomputing Center) via NSF ACCESS.

TEACHING EXPERIENCE

Graduate Teaching Assistant Fall 2023 – Spring 2026

Department of Mechanical Engineering, University of Utah

- ME EN 2450 — Numerical Methods (lab sections); ME EN 2455 — Programming for Modeling and Simulation; ME EN 2300 — Thermodynamics.
- Supported roughly 60 undergraduates per term; led in-person homework walkthroughs and two weekly office hours.
- Evaluated 20 undergraduate term projects and graded final examinations.

Lecturer Oct 2021 – Sep 2022

Daffodil International University, Dhaka, Bangladesh

- Authored the first version of the Outcome Based Education (OBE) curriculum for foundational mechanical and mechatronics engineering courses.
- Designed the mechatronics engineering drawing syllabus.

PEER-REVIEWED PUBLICATIONS

1. [Khalid Zobaid Adnan](#), Hao Zhou, and Tianli Feng, "Significant impact of $\text{Al}_{1-x}\text{Ga}_x\text{N}$ interlayer on GaN/AlN thermal boundary conductance," *Applied Physics Letters* **128**, 122201 (2026). doi:10.1063/5.0323350
2. [Khalid Zobaid Adnan](#), Tanvirul Abedien, and Tianli Feng, "Si/diamond thermal boundary conductance enhanced by SiN_x and amorphous carbon interlayers," *Physical Review Applied* **25**, 044013 (2026). doi:10.1103/84jf-svtg
3. Hao Zhou, [Khalid Zobaid Adnan](#), Wyatt Allen Jones, and Tianli Feng, "Thermal boundary conductance in standalone and nonstandalone GaN/AlN heterostructures predicted using machine-learning interatomic potentials," *Physical Review B* **112**, 235308 (2025). doi:10.1103/w7qp-tl6z
4. [Khalid Zobaid Adnan](#), Mahesh R. Neupane, and Tianli Feng, "Thermal boundary conductance of metal–diamond interfaces predicted by machine learning interatomic potentials," *International Journal of Heat and Mass Transfer* **235**, 126227 (2024). doi:10.1016/j.ijheatmasstransfer.2024.126227
5. [Khalid Zobaid Adnan](#) and Tianli Feng, "Thermal boundary conductance and thermal conductivity strongly depend on nearby environment," *Physical Review B* **109**, 245302 (2024). doi:10.1103/PhysRevB.109.245302

CONFERENCE PRESENTATIONS

- "Thermal boundary conductance of metal–diamond interfaces predicted by machine learning interatomic potentials," conference paper and presentation, ASME Summer Heat Transfer Conference (SHTC), 2025.
- "Thermal boundary conductance and thermal conductivity strongly depend on nearby environment," poster, ASME Summer Heat Transfer Conference (SHTC), 2025.

RESEARCH SOFTWARE

- **MLIPs for interfaced materials** — public repository of custom-trained DeePMD and MTP potentials for heterogeneous interfaces including Si/diamond and Au/diamond, packaged for LAMMPS thermal boundary conductance runs.
- **JAX-MLIP** — harmonic-constrained interatomic potentials in JAX that learn force-constant structure from displacement data.
- **Autoencoder DMD** — autoencoders and dynamic mode decomposition for extracting phonon modes and forces from ab initio molecular-dynamics trajectories.

SELECTED COURSEWORK PROJECTS

Heat exchanger and thermal system design (2025), Python and EES optimization models for solar/evaporation-pond and brine-reheat systems · **Radiation heat transfer solver** (2024), iterative solver for heat-flux distributions in participating media · **Amorphous phonon mean-free-path prediction** (2024), grey-model code for thermal properties of amorphous structures.

TECHNICAL SKILLS

Simulation: LAMMPS, VASP, Quantum ESPRESSO, GPUMD, Phonopy, COMSOL, ANSYS

Machine learning: DeePMD-kit, MTP (MLIP-2), FitSNAP, PyTorch, JAX

Programming: Python, MATLAB, C++, C; SLURM and multi-GPU HPC workflows

Engineering tools: EES, OriginLab, SolidWorks